Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *French Hospital Clinical NLP — Metropolitan France (DrBenchmark)*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output Ontology was flagged HIGH priority in elicitation due to the structural mismatch between the deployment's confidence-scored multi-label outputs with uncertainty flagging and the benchmark's standard classification metrics. The benchmark provides rich label ontologies — POS tags (CAS 31, ESSAI 36) [Q101, Q102], NER schemes (QUAERO 10 UMLS Semantic Groups [Q103, Q35], E3C clinical/temporal [Q104], DEFT-2021 14 entity types [Q108], Mantra-GSC [Q106]), DEFT-2021 23 MeSH axes [Q27], DiaMed 22 ICD-10 chapters [Q45, Q107], MorFITT 12 specialties [Q105]. DEFT-2021's NER scheme covers exactly the entity types the deployment needs (PATHOLOGY, ANATOMY, TREATMENT, DOSAGE, DURATION, FREQUENCY, MOMENT, DATE) [Q108, DEFT2021-D3, DEFT2021-D4, DEFT2021-D5]. However, multiple structural mismatches harm output ontology validity: (1) CRITICAL — no benchmark task has confidence-scored, ranked-candidate, or uncertainty-flagging outputs; DEFT-2021's specialities_one_hot is strictly binary 0/1 (DEFT2021-D11), DiaMed is single-label per case (DIAMED-D10), and DEFT-2020 averages annotator scores into a single float (DEFT2020-D16); (2) ICD-10 chapter granularity in DiaMed (22 categories) is far coarser than operational PMSI/GHM codes derived from CIM-10 FR (DIAMED-D11) [WEB-2]; (3) several datasets have output taxonomies orthogonal to the deployment's pathology classification function — CAS/ESSAI measure negation/speculation modality (CAS-D19, ESSAI-D9), FrenchMedMCQA measures answer-set selection (FRENCHMEDMCQA-D1), DEFT-2020 measures sentence similarity (DEFT2020-D14, DEFT2020-D15); (4) UMLS Semantic Group NER provides no ICD-10 sub-typing (MANTRAGSC-D10); (5) E3C French_clinical uses only binary CLINENTITY annotation, underspecifying multi-type pathology distinction (E3C-D14); (6) QUAERO and Mantra-GSC contain no temporal entity labels at all (QUAERO-D13, QUAERO-D14, MANTRAGSC-D9); (7) PxCorpus NER lacks pathology/diagnosis entities (PXCORPUS-D16). The benchmark explicitly notes that even specialized models perform poorly on MLC and STS [Q82, Q86, Q87], directly implicating the deployment's most challenging function.

ID	Evidence
Q27	"The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH)." (p.3)
Q35	"10 entity categories corresponding to the UMLS Semantic Groups (Lindberg et al., 1993) were annotated (more detail in Appendix B.3)." (p.3)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q82	"Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS." (p.8)
Q86	"We have observed that several biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models." (p.9)
Q87	"We postulate that the models examined in this study, here state-of-the-art MLMs, may not be the most effective choices for specific tasks such as question-answering or multi-label classification." (p.9)
Q101	"INT, PRO:DEM, VER:impf, VER:ppre, PRP:det, KON, VER:pper, PRP, PRO:IND, VER:simp, VER:con, SENT, VER:futu, PRO:PER, VER:infi, ADJ, NAM, NUM, PUN:cit, PRO:REL, VER:subi, ABR, NOM, VER:pres, DET:ART, VER:cond, VER:subp, DET:POS, ADV, SYM and PUN." (p.14)

Q102	“INT, PRO:POS, PRP, SENT, PRO, ABR, VER:pres, KON, SYM, DET:POS, VER:, PRO:IND, NAM, ADV, PRO:DEM, NN, PRO:PER, VER:pper, VER:ppre, PUN, VER:simp, PREF, NUM, VER:futu, NOM, VER:impf, VER:subp, VER:infi, DET:ART, PUN:cit, ADJ, PRP:det, PRO:REL, VER:cond and VER:subi.” (p.14)
Q103	“O, GEOG, PHEN, DISO, ANAT, OBJC, PHYS, PROC, DEVI, CHEM and LIVB” (p.14)
Q104	“Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML” (p.14)
Q105	“microbiology, etiology, virology, physiology, immunology, parasitology, genetics, chemistry, veterinary, surgery, pharmacology and psychology” (p.14)
Q106	“Medline: ANAT, PROC, CHEM, PHYS, GEOG, DEVI, LIVB, OBJC, DISO, PHEN and O. EMEA and Patents: ANAT, PROC, CHEM, PHYS, DEVI, LIVB, OBJC, DISO, PHEN and O.” (p.14)
Q107	“Multi-label Classification: immunitaire (immunology), endocriniennes (endocrinology), blessures (injury), chimiques (chemicals), etatsosy (signs and symptoms), nutritionnelles (nutrition), infections (infections), virales (virology), parasitaires (parasitology), tumeur (oncology), osteomusculaires (osteomuscular disorders), stomatognathique (stomatology), digestif (digestive system disorders), respiratoire (respiratory system disorders), ORL (otorhinolaryngologic diseases), nerveux (nervous system disorders), oeil (eye diseases), homme (male genital diseases), femme (female genital diseases), cardiovasculaires (cardiology), hemopathies (hemic and lymphatic diseases), genetique (genertic disorders) and peau (dermatology).” (p.14)
Q108	“Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE” (p.14)
FRENCHMED MCQA-D1	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: Une population de lymphocytes>30% — Clinical hematology question in formal medical French
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
DEFT2021-D4	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Multiple entity types co-occurring in a single sentence, confirming NER scheme coverage
DEFT2021-D5	Au jour 42 , une deuxième désensibilisation a eu lieu — Temporal anchor (relative day) labeled, confirming temporal entity extraction alignment
ESSAI-D9	Essai clinique d'immunothérapie évaluant l'association du galunisertib (un facteur de croissance transformant β), avec l'anticorps anti-PD-L1 durvalumab (MEDI4736), dans le cancer du pancréas métastatique — Rich pathology content labeled only for negation/speculation, not ICD-10 — output ontology mismatch
MANTRAGSC-D9	Traitement En cas d' ingestion récente, l' éventualité de provoquer un vomissement et d' effectuer un lavage gastrique devra être considérée. — Temporal word "récente" present but no temporal NER label exists in Mantra-GSC scheme
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
MANTRAGSC-D10	Nouveautés dans la dysplasie ventriculaire droite arythmogène. — Single DISO class applied to cardiovascular disorder; no ICD-10 subcategory distinction
DEFT2021-D11	[1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0] — Strictly binary 0/1 label vector — no confidence or soft-label dimension present
DIAMED-D11	Maladie de Kaposi à localisation broncho-pulmonaire révélant une infection VIH — Chapter-level label only; PMSI granular codes (e.g., B21.x) not evaluated
QUAERO-D13	La durée du traitement est allée d'une heure de perfusion en bolus à une utilisation continue de plus de 6 ans. — Temporal expressions present but unlabeled; confirms absence of temporal NER
DEFT2020-D14	Fields: id, moy, vote, scores, source, cible — Continuous similarity scoring schema; no entity labels, no ICD-10 classification outputs
E3C-D14	"acute respiratory distress syndrome" tagged as single CLINENTITY span, no sub-type — Coarse two-class annotation lacks pathology/anatomy/treatment distinction needed by deployment
QUAERO-D14	La dose peut être augmentée par intervalles de 1 à 2 jours, voire plus… — Another unlabeled temporal phrase; reinforces temporal NER gap
DEFT2020-D15	Field: correct_cible (0/1/2) — Sentence selection index; not a pathology classification or ICD-10 label
DEFT2020-D16	Après inspection, elles ont toutes été exclues. — Annotator disagreement averaged away in scoring; calibration signal lost
PXCORPUS-D16	NER tags: dose, form, frequency, duration, symptom only — No PATHOLOGY/DIAGNOSIS entity in tag scheme; ICD-10 mapping untestable
CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)

Strengths

- DEFT-2021 NER ontology covers all entity types the deployment requires: PATHOLOGY, ANATOMY, TREATMENT, DOSAGE, DURATION, FREQUENCY, MOMENT, DATE [Q108, DEFT2021-D3, DEFT2021-D4]
- DiaMed provides ICD-10 chapter-level labels aligned with the deployment's classification ontology, the only direct ICD-10 alignment in the benchmark (DIAMED-D1, DIAMED-D2)
- DEFT-2021 23 MeSH axes support multi-label structure with up to 7 concurrent labels per case (DEFT2021-D1, DEFT2021-D2)
- Benchmark's own findings flag MLC/STS as poor-performance areas [Q82, Q86] — useful diagnostic signal for deployment risk

ID	Evidence
Q82	“Upon analyzing the average performance by task category, it becomes evident that the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS.” (p.8)
Q86	“We have observed that several biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models.” (p.9)
Q108	“Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE” (p.14)
DIAMED-D1	Le test rapide VIH était positif, confirmé par la sérologie VIH avec un taux de CD4 à 27/mm3 — Confirmed HIV infection with CD4 count — supports infectious disease label
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
DIAMED-D2	L'examen anatomopathologique de la pièce opératoire montrait une prolifération tumorale faite de travées, de cordons — Histopathological finding of tumoral proliferation — confirms neoplasm label
DEFT2021-D2	Nous rapportons le cas d'un garçon de huit ans [...] microangiopathie thrombotique secondaire à un syndrome hémolytique urémique (SHU) atypique — 7-label pediatric multi-morbid case illustrating complex label overlap
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
DEFT2021-D4	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Multiple entity types co-occurring in a single sentence, confirming NER scheme coverage

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories partially align: NER tag schemes (DEFT-2021, E3C temporal) cover deployment-needed entity types [Q108, Q104]; ICD-10 chapter labels in DiaMed align coarsely with PMSI mapping [Q45]; POS tag inventories match standard French biomedical morphology [Q101, Q102].

OO-2: Missing categories: (1) confidence scores / ranked candidates / uncertainty flags — absent across all 20 tasks [Q56, DEFT2021-D11, DIAMED-D10]; (2) PMSI-granular codes — DiaMed only at chapter level (DIAMED-D11) [WEB-2]; (3) tropical/DOM-TOM-specific pathology subcategories — absent from French subsets (E3C-D4, E3C-D13); (4) pathology entities in PxCorpus NER (PXCORPUS-D16); (5) temporal labels in QUAERO and Mantra-GSC (QUAERO-D13, MANTRAGSC-D9).

OO-3: MorFITT's 12 medical specialty labels (microbiology, surgery, pharmacology) encode research-specialty routing rather than ICD-10 diagnostic axes, and do not reflect French hospital coding conventions (MORFITT-D11, MORFITT-D21). CAS/ESSAI's negation/speculation modality measures linguistic phenomena, not pathology classification (CAS-D19, ESSAI-D9). DEFT-2020 STS continuous similarity score has no diagnostic correspondence (DEFT2020-D14).

OO-4: A stakeholder-driven taxonomy redesign would be needed to introduce: (a) calibrated confidence scoring layer over multi-label outputs; (b) PMSI/GHM code-level granularity beyond ICD-10 chapter; (c) ranked-candidate output format with uncertainty thresholds. The deployment's Silver Standard framing requires these for clinician adjudication.

OO-5: Structural validity is harmed by absence of confidence/uncertainty output structure [Q56, DEFT2021-D11]. Content validity is harmed by orthogonal classification labels (negation/speculation in CAS/ESSAI; STS continuous in DEFT-2020/CLISTER; pharmacy MCQA in FrenchMedMCQA) and by chapter-level rather than PMSI-level granularity [WEB-2]. The benchmark itself observes that 'biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models' [Q86] and 'state-of-the-art MLMs may not be the most effective choices for specific tasks such as question-answering or multi-label classification' [Q87].

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Q56	“The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions.” (p.6)
Q86	“We have observed that several biomedical tasks in DrBenchmark exhibit relatively poor performance, even when utilizing specialized biomedical models.” (p.9)
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Q102	“INT, PRO:POS, PRP, SENT, PRO, ABR, VER:pres, KON, SYM, DET:POS, VER:, PRO:IND, NAM, ADV, PRO:DEM, NN, PRO:PER, VER:pper, VER:ppre, PUN, VER:simp, PREF, NUM, VER:futu, NOM, VER:impf, VER:subp, VER:infi, DET:ART, PUN:cit, ADJ, PRP:det, PRO:REL, VER:cond and VER:subi.” (p.14)
Q104	“Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML” (p.14)
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DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
E3C-D4	The presence of terminal spine confirmed Schistosoma haematobium and a diagnosis of schistosomiasis was made. — Tropical parasitic pathology in English config — confirms tropical content structurally absent from French subset
MANTRAGSC-D9	Traitement En cas d' ingestion récente, l' éventualité de provoquer un vomissement et d' effectuer un lavage gastrique devra être considérée. — Temporal word "récente" present but no temporal NER label exists in Mantra-GSC scheme
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DEFT2021-D11	[1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0] — Strictly binary 0/1 label vector — no confidence or soft-label dimension present
DIAMED-D11	Maladie de Kaposi à localisation broncho-pulmonaire révélant une infection VIH — Chapter-level label only; PMSI granular codes (e.g., B21.x) not evaluated
MORFITT-D11	Nous rapportons le cas d'une patiente de 71 ans ayant développé un pseudomyxome péritonéal secondaire à un adénocarcinome appendiculaire. — Clinical case abstract with surgery+etiology labels; labels do not map to ICD-10 codes

QUAERO-D13	La durée du traitement est allée d'une heure de perfusion en bolus à une utilisation continue de plus de 6 ans. — Temporal expressions present but unlabeled; confirms absence of temporal NER
E3C-D13	a diagnosis of schistosomiasis was made — Tropical disease in English config only; absent from French subset — confirms DOM-TOM pathology gap
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PXCORPUS-D16	NER tags: dose, form, frequency, duration, symptom only — No PATHOLOGY/DIAGNOSIS entity in tag scheme; ICD-10 mapping untestable
CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
MORFITT-D21	Bloquer le complément, notamment l'axe C5a-C5aR1, par des thérapies spécifiques représente un espoir thérapeutique dans les formes les plus sévères de la maladie. — COVID-19 immunology; three specialty labels that don't resolve to ICD-10 output
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)

Information Gaps

- Whether the 23 MeSH axes in DEFT-2021 map systematically to PMSI Catégories Majeures groupings

Requires Expert Verification

- Calibration thresholds appropriate for the deployment's uncertainty-flagging behavior; benchmark provides no protocol for setting these

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Remediation

Gap 1: No benchmark task evaluates confidence-scored multi-label outputs, uncertainty flagging, or ranked-candidate quality — the deployment's central output behavior is structurally unmeasurable; ICD-10 chapter granularity is coarser than operational PMSI/GHM codes

Recommendation: Extend the DrBenchmark protocol on DEFT-2021 and DiaMed with calibration metrics (Expected Calibration Error, Brier score, per-label AUROC, top-k ranked-candidate F1). Build a supplementary PMSI-code-level evaluation set using a hospital-internal stratified sample mapped to CIM-10 FR / GHM groupings per ATIH 2024 guidance [WEB-2].

ID	Evidence
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)